Domino Computer



Intro

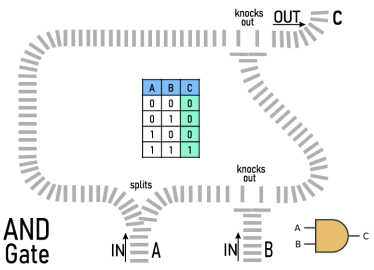
Computers perform calculations using binary numbers and logic gates. In this activity, the electrical circuit is replaced with domino blocks to show that the same logical operations can be demonstrated with tangible objects. A network of dominoes can add numbers in the same way a computer processor does, but much more slowly.

The aim of this activity is to demonstrate how simple reactions, such as falling dominoes, can be organised to perform mathematical calculations. It also helps learners understand the basic working of logic gates in computers.

Material Required

Domino Blocks
Logic Gate Flex
A flat surface for construction.





Learning Goals

- How computers represent information using just two states: ON and OFF
- What logic gates are and how the three simplest gates (OR, AND, XOR) work
- How binary numbers work and why they're perfect for computers
- How logic gates combine to create circuits that perform calculations
- How a network of dominoes can model the way a real computer adds numbers

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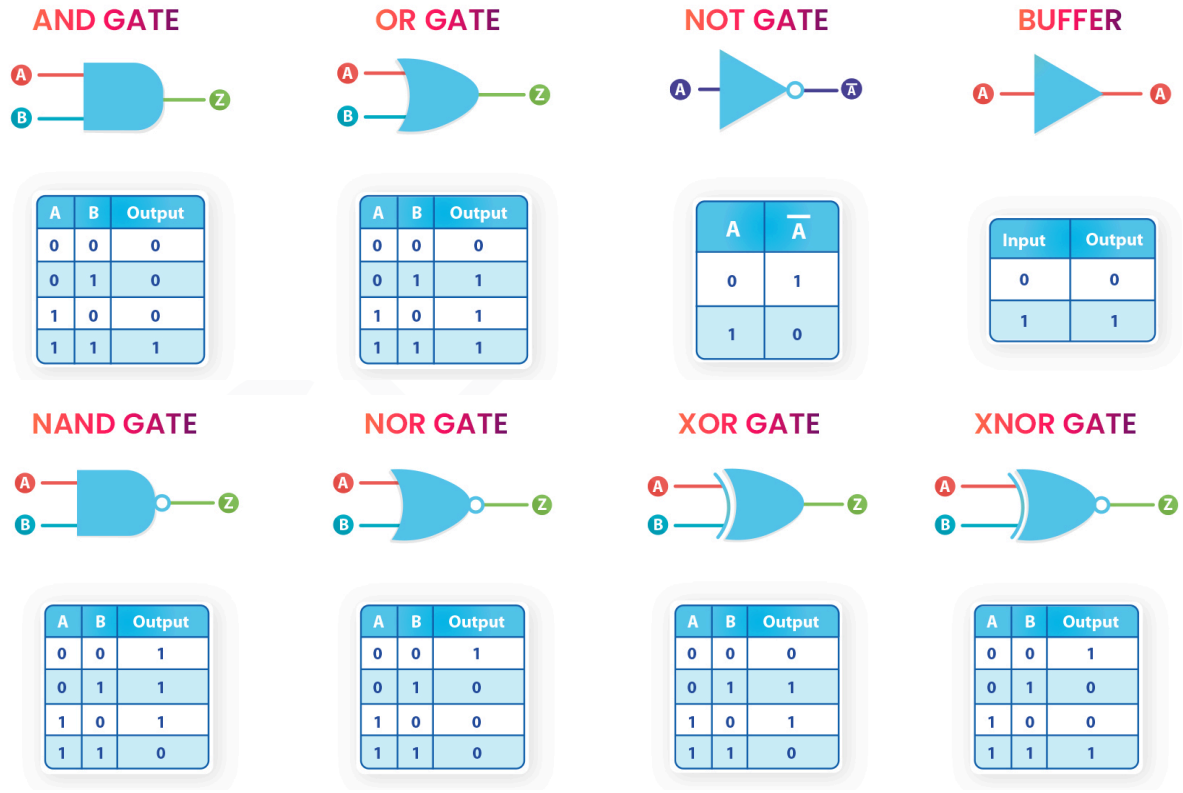
Instructions

Computers do not work with numbers in the same way humans do. Instead, they break everything into tiny signals that are either present or absent, on or off, 1 or 0. Computers are made up of logic gates. Logic gates take in information and output different information depending on the type of gate.

A logic gate is a decision-making unit inside a computer. It takes one or more inputs and gives an output according to a fixed rule. In our domino model, the input is whether a domino block is falling, and the output is whether another domino block falls in response.

In computers, information is represented as electrical signals that can be ON or OFF (high or low voltage). In our domino computer, the information is whether a chain of domino blocks is falling.

There are many different types of logic gates; some are shown.



A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

A	$\overline{A}$
0	1
1	0

Input	Output
0	0
1	1

A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	0

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Combinations of different gates allow the computer to perform complicated calculations, and the output from one gate can be connected to other gates.

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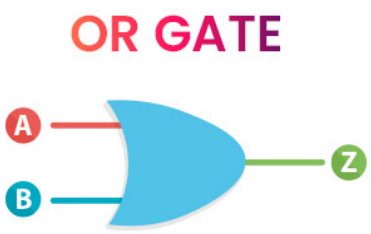
By assembling enough domino blocks into just the right sequence, we can create a structure out of those dominoes that acts as a computer circuit: representing numbers, calculating sums, and ultimately performing any computation.

Computers are powered by electricity. If we replace the current flowing through a wire with dominoes falling along a path, we can model it using dominoes. We represent a 0, or the absence of electricity, with a standing domino, and we represent a 1, or the presence of electricity, by toppling a domino and starting a chain reaction.

Standing domino = 0 (OFF) - no signal
Falling domino = 1 (ON) - signal is flowing

Ultimately, though, computers need logic: some system to convert inputs to outputs. Computer circuits consist of logic gates, which are small electronic units that perform simple logical functions.

OR Gate
The OR gate is a logic gate that passes a signal if one or more of its inputs are on. The symbol for an OR gate is shown below, along with a table of results for all possible input combinations.



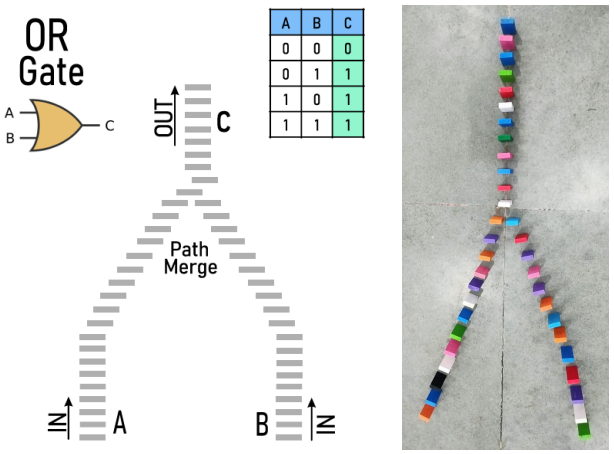
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

When we use the word 'OR' in any conversation, we usually mean "one thing or the other thing, but not both". It's important to remember that the logic gate called OR means "one or the other or both".

To make an OR gate from domino blocks, we need two input chains, either of which will set off the output chain of domino blocks.

A domino block OR Gate looks like this:

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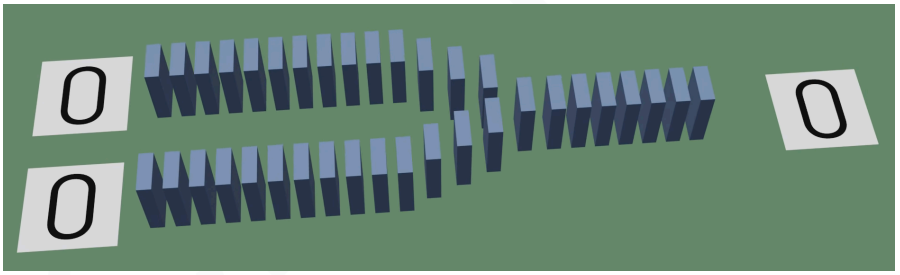


An OR gate is one of the simplest logic gates. It has two binary inputs; if both inputs are 0, the output is 0. But if the first or second input is a 1, or if both are 1s, then the output is a 1.

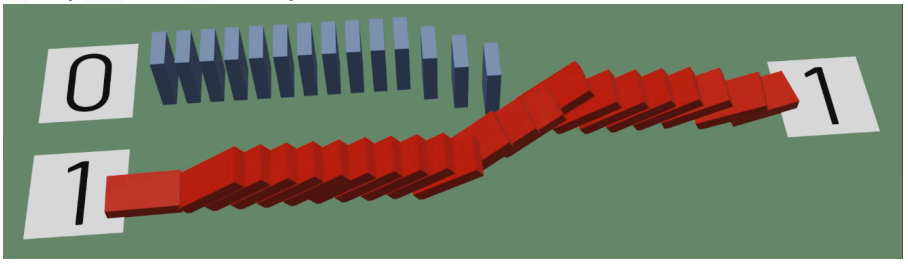
In domino terms, this means we have two domino inputs, and we want the output to fall if either input is pushed.

The input domino blocks are placed such that when you push either one / both of them, the output domino block will fall. If neither input is pushed, the output remains in the off (0) state.

If no input is pushed, the output is 0.

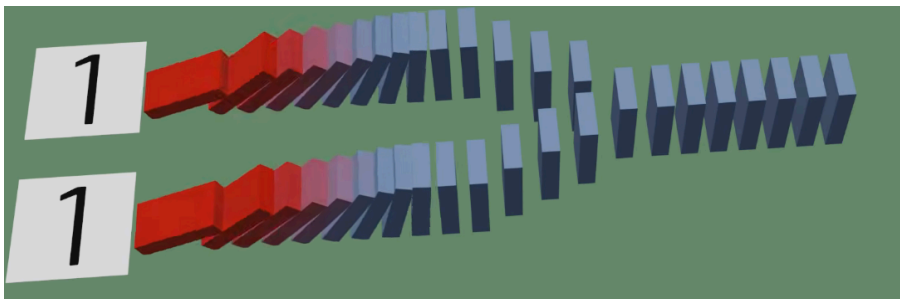


If either input is pushed, the output is 1.

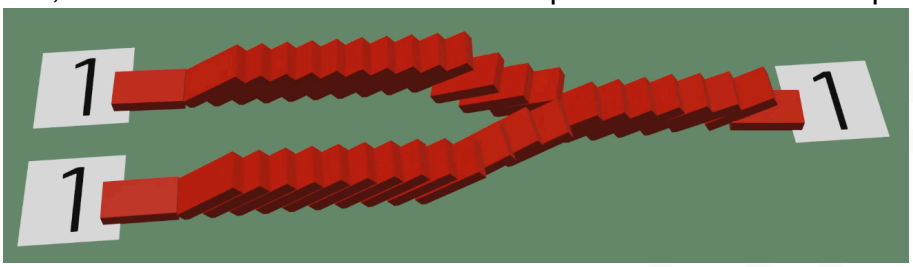


And if both inputs are pushed, will the output domino fall?

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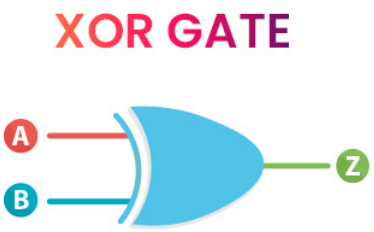


Answer: Yes, because OR allows one or both inputs to activate the output.



XOR Gate

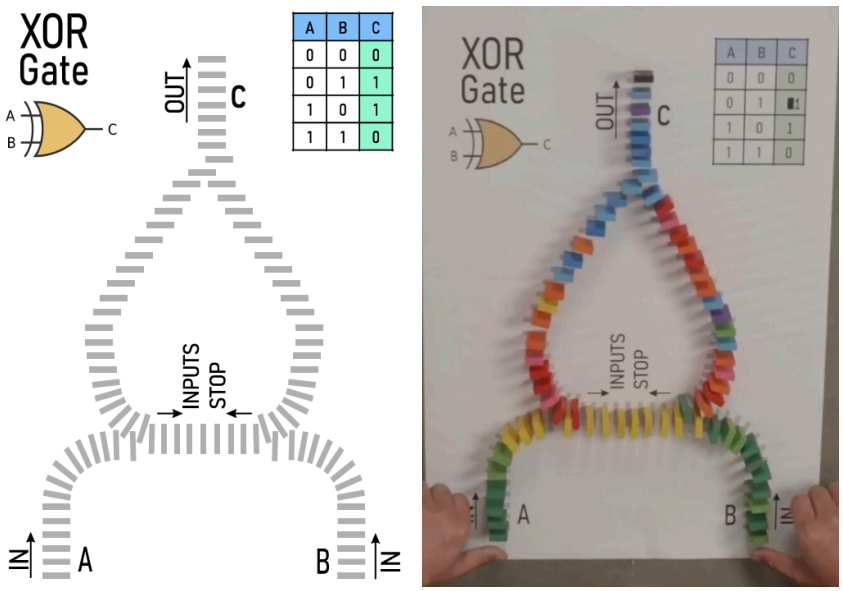
The XOR gate, EXCLUSIVE OR, is a logic gate which passes on a signal if exactly one of the two inputs is on. This is different from the OR gate, which will still pass a signal if both inputs are 1. The difference is that with exclusive OR, the two inputs are mutually exclusive. In other words, if both inputs are 1, then the two inputs need to cancel out somehow and produce a 0 instead of a 1. The symbol for an XOR gate is shown, along with a table of results for all possible input combinations.



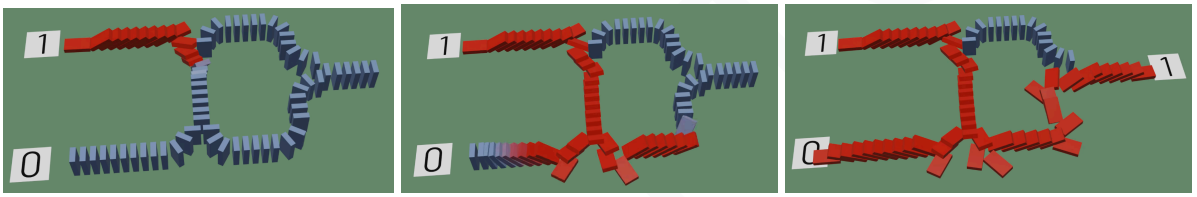
A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0

To make an XOR gate from domino blocks, we need two input chains, either of which will set off the output chain of cards, but not both. This can be achieved by having the two inputs pass through the same section of blocks; if they're both running, they will stop each other. This can be achieved with a gate like this:

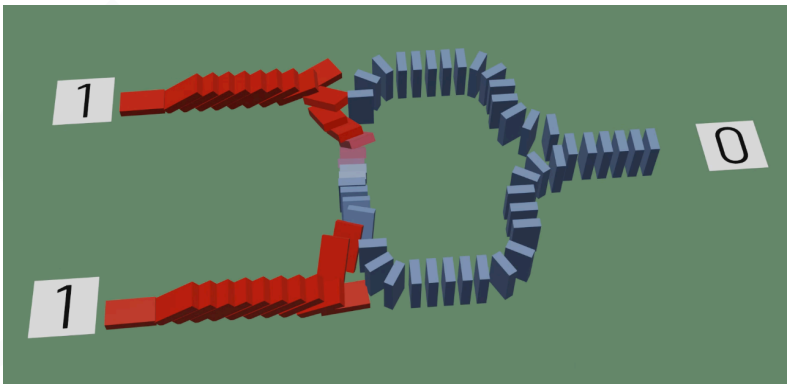
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If only one input is pushed, the dominoes travel around the corresponding curve, and the output is a 1.



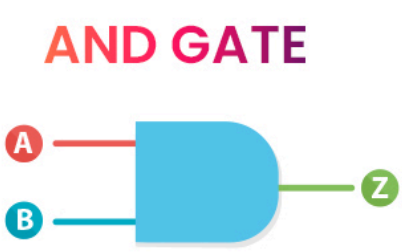
But if both inputs are pushed, the two paths collide and stop midway. As a result, the output falls over when either input is pushed, but not when both inputs are pushed.



AND Gate

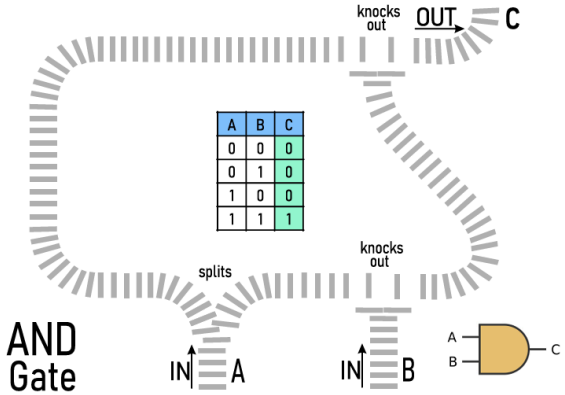
The AND gate is a logic gate that passes a signal only when both inputs are on. The symbol for an AND gate is shown, along with a table of results for all possible input combinations.

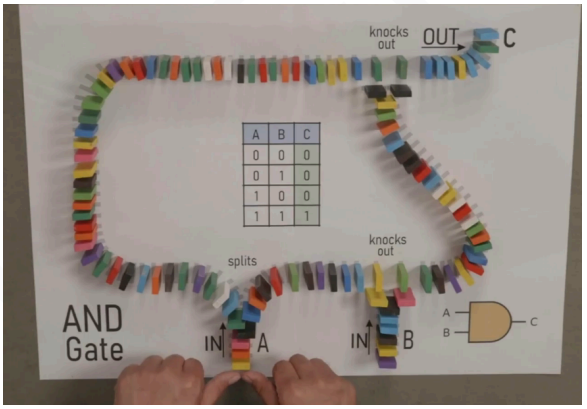
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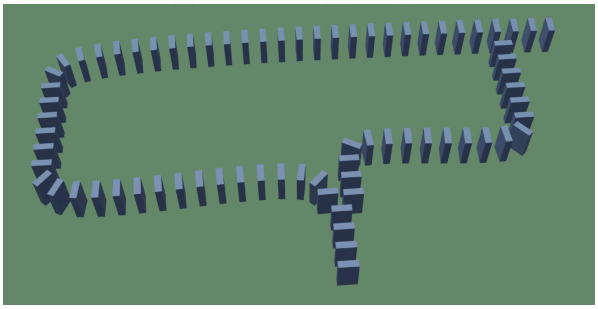
A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

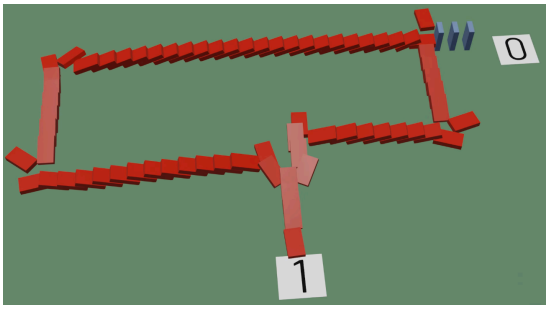
The AND gate takes two inputs and produces a 1 only when both inputs are a 1. To make an AND gate from domino blocks, we need the output chain to fall only when both input chains fall. Representing that kind of computation with dominoes feels tricky. How could we design a domino output that falls only when both inputs fall? Here, we can again take advantage of the falling domino's ability to halt the progress of other dominoes. One way to do this is to split an input chain so it can knockout itself, and use the second input to interrupt the first input's knockout chain.





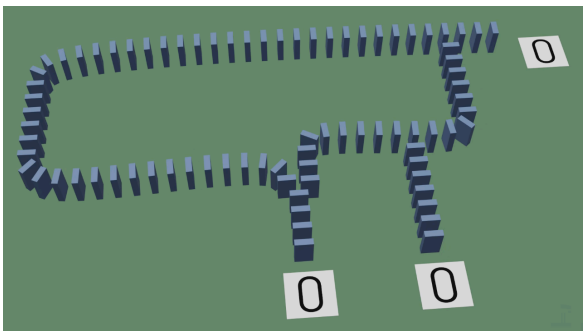
We start with one input that has a relatively long path to the output. We then add an extra path from that input that can interrupt the longer path. If the input is pushed, the shorter domino path will interrupt the longer one, and the output at the end will not fall over.

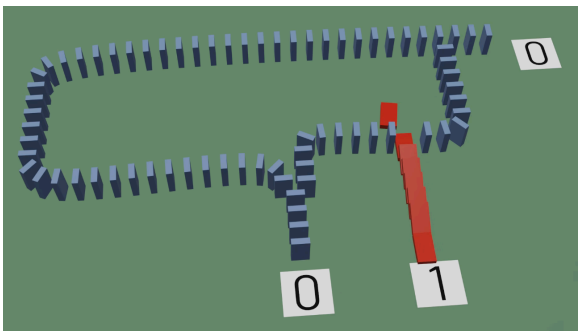




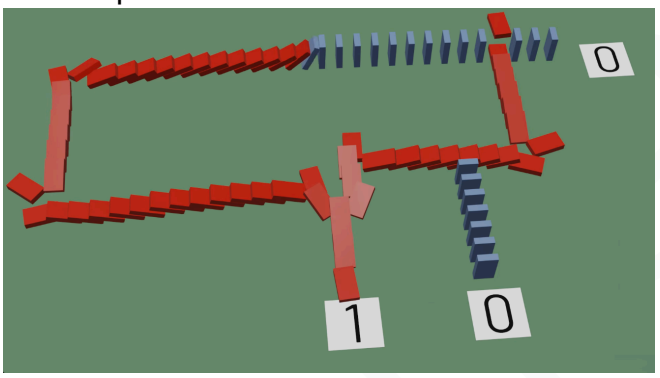
To create an AND gate, we just need to add a second input that interrupts the knockout path. If just the second input is pushed, nothing happens to the output.

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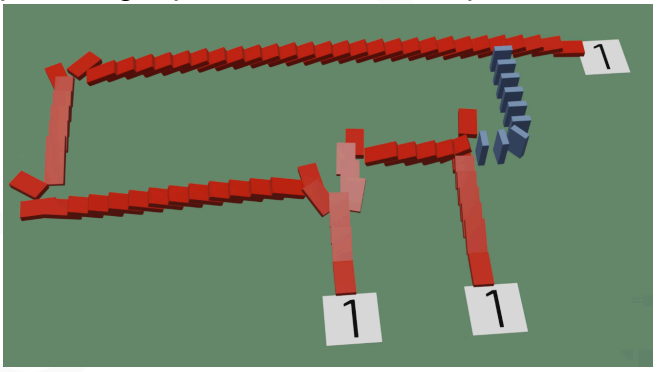




If only the first input is pushed, the interrupt path blocks the longer path, and nothing happens to the output either.



But if both inputs are pushed, the second input interrupts the knockout path, allowing the first input's longer path to reach the output.



Can we build NOT and NAND gates out of domino blocks?

Binary Numbers

Since computers store information using electrical signals, there are only two states they can represent: ON or OFF. Similarly, the domino computer uses chains of dominoes that are either falling or not falling to transmit information.

This means that if we are going to use them to handle numbers, we need a way to store them using only two values. This is called BINARY.

A number like **45** is made up of different digits, each representing a different value.

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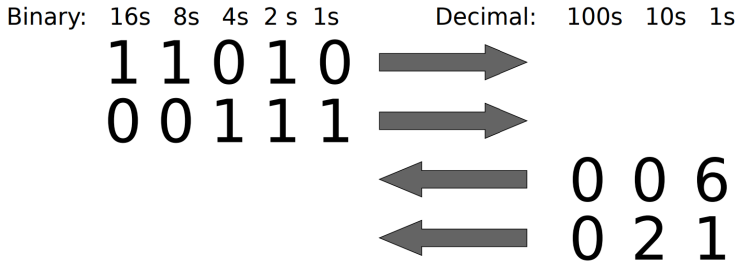
45 = (4 * 10) + (5 * 1)

The right-hand digit is ones, the next digit is tens, the third digit from the right is hundreds, and so on. This means we need the digits 0-9 to record every possible value. We humans count using 10 digits (0–9). This is called decimal, or base-10. Why 10? Probably because we have 10 fingers.

But computers don't have fingers. They have logic gates that are either ON or OFF. So computers use binary, or base 2, which uses only two digits: 0 and 1. In binary, instead of using 1, 10, 100, 1000... as our values, which are powers of 10, we instead use powers of 2. These are 1, 2, 4, 8, 16, 32... (doubling every time). With these, we can record any number, and we will only ever need 0 or 1 in each column.

101101 = 45 = (1 * 32) + (0 * 16) + (1 * 8) + (1 * 4) + (0 * 2) + (1 * 1)

Practice converting numbers between binary and decimal!



How many binary digits are needed to represent the number 1947?

Binary numbers can be added together in exactly the same way as decimal numbers. The domino computer takes two numbers written in binary, which we provide by filling gaps in the input chains, and the output chains will contain their sum.

From Logic Gates to Arithmetic

Individual logic gates can be connected to one another. The output of one gate becomes the input to another gate. When you combine enough gates in just the right way, they can add numbers together.

Can we build a domino circuit that adds two 1-bit binary numbers?

What happens when we add in binary?

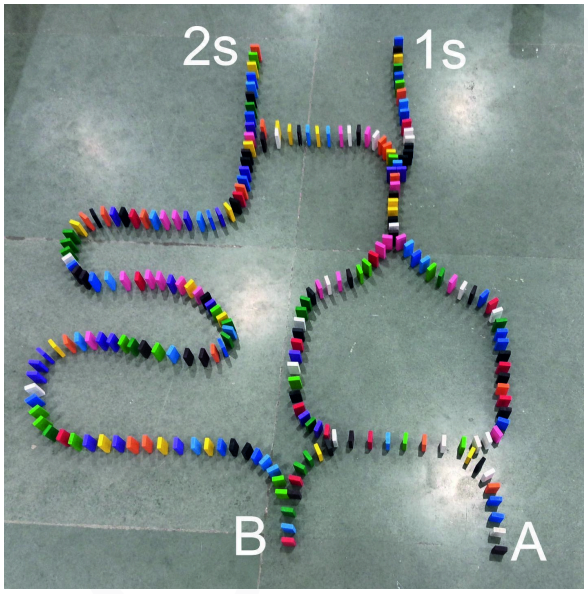
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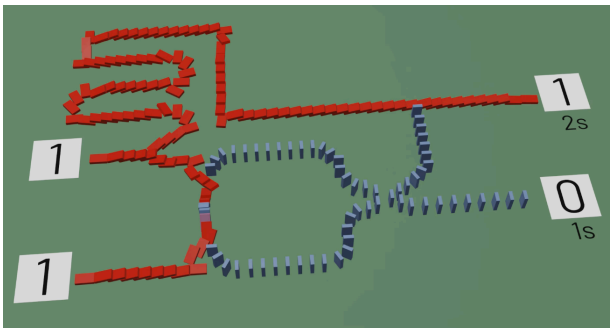
0 + 0 = 0
0 + 1 = 1
1 + 0 = 1
1 + 1 = 10 (which is 2 in decimal)

Notice that last one: 1 + 1 = 10. The result is two digits! The "1" goes into the next column (this is called a "carry").

The domino-block circuitry for adding 2-bit numbers A & B is called a half adder.

A	B	2s	1s
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0



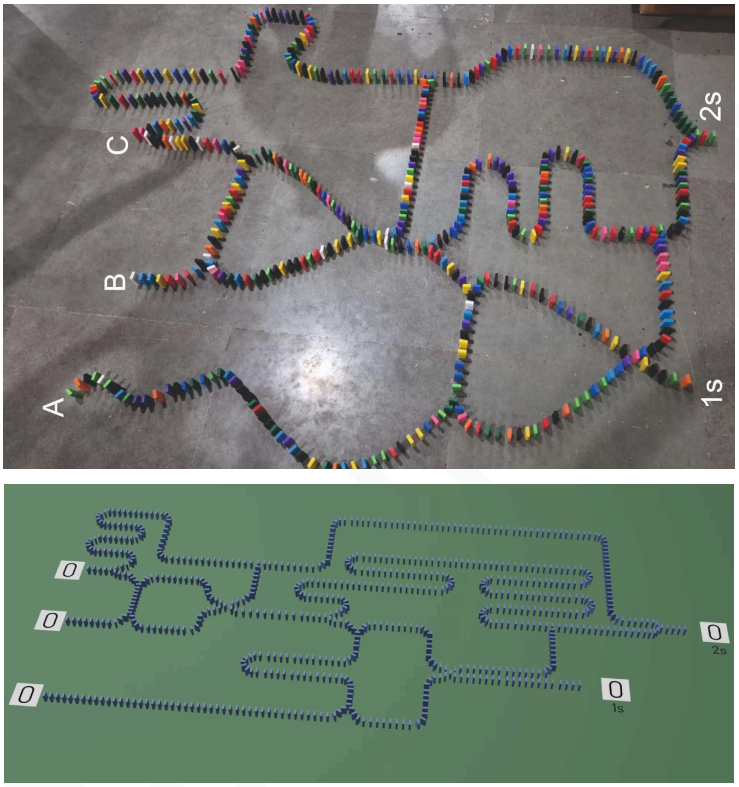


If you made the half adder, then make the full adder by connecting two half adders.

A	B	C	2s	1s
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1

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0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1



Basically, a sequence of dominoes organised into logic gates: those logic gates combine to form half adders, which combine to create full adders, which, when put in sequence, can add any two binary numbers by very precisely determining which lines of domino blocks should fall and which shouldn't.

Computer circuits combine to perform more and more sophisticated computations, at speeds much faster than our domino computer can compute, of course, but using fundamentally the same logic.

The domino computer turns an invisible idea into something we can see and test. By watching domino chains represent signals, one begins to understand how computers make decisions, store information, and perform calculations. Although computers use electricity instead of dominoes, the underlying logic is the same. This activity

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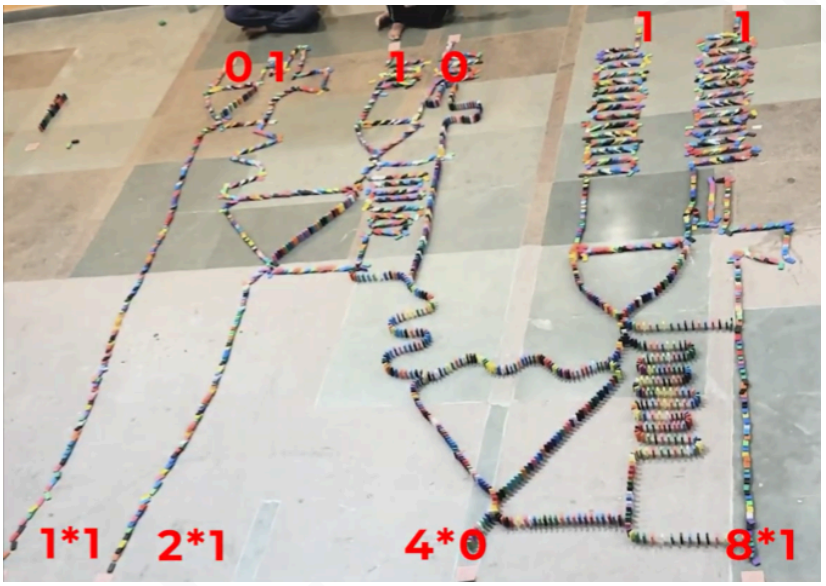
gives an early and memorable introduction to computational thinking, binary numbers, and the role of logic gates in modern computing.

Fact

In October 2012, Matt Parker and the Think Maths team built a domino computer at the Manchester Science Festival. A working binary adder made of 10,000 dominoes.

It took 12 people six hours to carefully balance every domino block. When they push the dominoes, the chain reaction cascades through, calculating the sum of two 4-bit binary numbers and producing the 5-bit result.

At IIT Gandhinagar's Centre for Creative Learning, we built a version that adds two 3-bit binary numbers to produce a 4-bit result (e.g., 5 + 6 = 11).



Video Link: https://tinyurl.com/dominotryout

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